Granulomatus dermatosis

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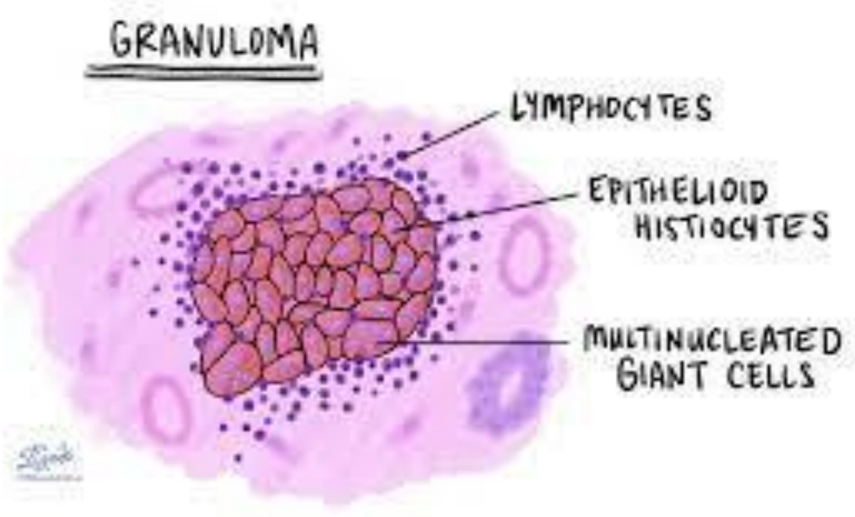
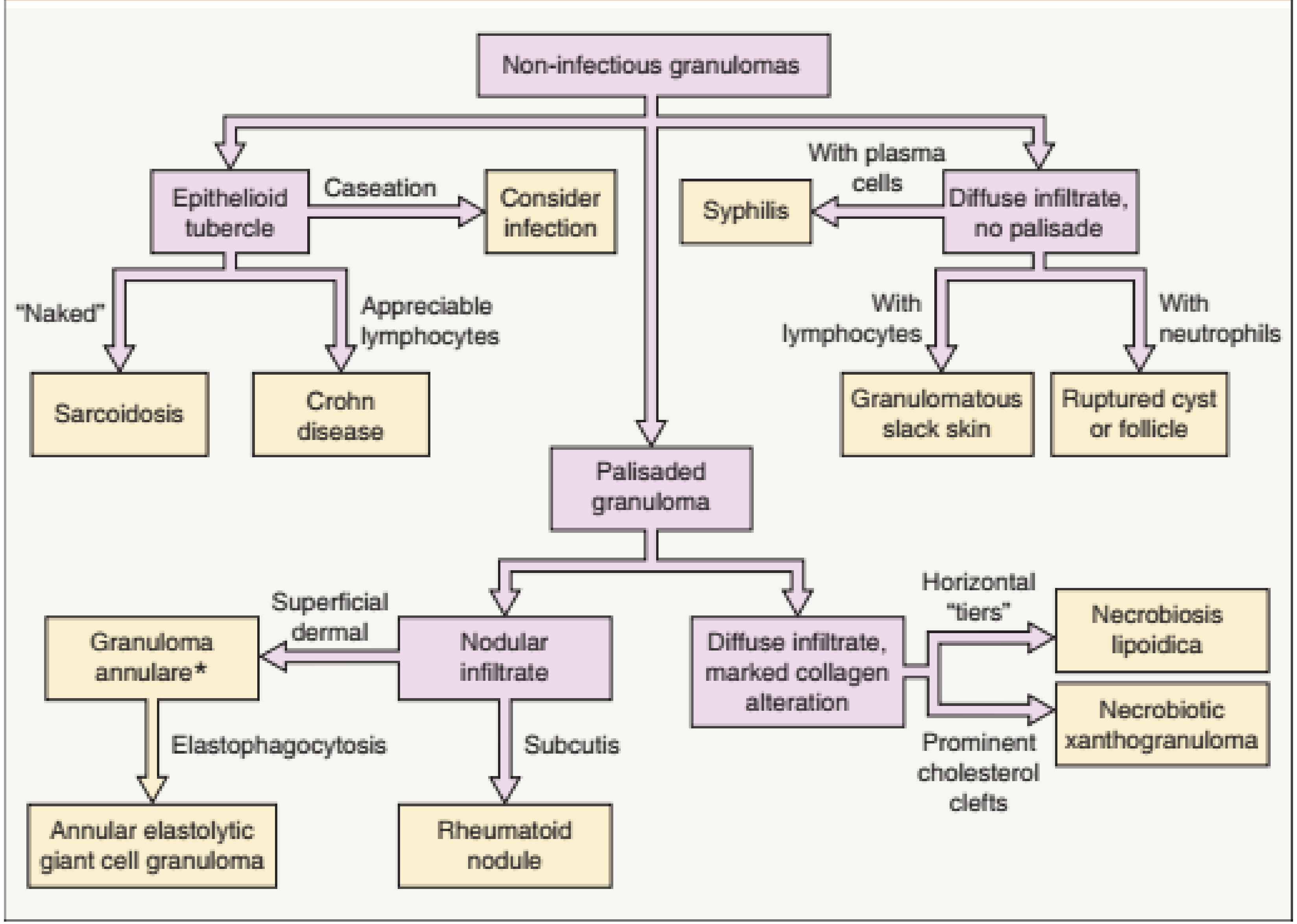
granuloma

- definition:
- chronic inflammatory condition of skin in which Histeocytes are predominant cells.



NON-INFECTIOUS GRANULOMAS: ALGORITHM FOR HISTOLOGIC DIAGNOSIS



TYPES OF GRANULOMA

types	charactrestics	dermatosis
tuberculoid	Langhans gaint cells with dense rim of lymphocytes +_ caseation	TB-TL-LMDF-leishmania
sarcoidal	asteroid ,shuman body Sparse lymphocyte (naked granuloma)	Sacoidosis, sarcoidal reaction
palisading	Palisade like around necrobiosis	GA-NBL-RN-AEGCG-NXG
Foreign body granuloma		
suppurative	With neotrophilis	INFECETIOS AGENT

CLINICAL FEATURES OF THE MAJOR NON-INFECTIOUS GRANULOMATOUS DERMATITIDES						
	Sarcoidosis*	Classic granuloma annulare†	AEGCG	Necrobiosis lipoidica	Cutaneous Crohn disease	Rheumatoid nodule
Average age (years)	25–35, 45–65	<30	50–70	30	35	40–50
Sex predilection	Female	Female	Female	Female	Female	Male‡
Racial/ethnic predilection in US	African-American	None	Caucasian	None	Ashkenazi Jews	None
Sites	Symmetric on face, neck, upper trunk, extremities	Hands, feet, extensor aspects of extremities	Face, neck, forearms (sites of chronic sun exposure)	Anterior and lateral aspects of shins	Anogenital region, buttocks, lower > upper extremities	Juxta-articular areas, especially elbows and hands
Appearance	Protean; most commonly red– brown or violaceous papules and plaques	Papules coalescing into annular plaques	Annular plaques	Plaques with elevated borders, telangiectasias centrally	Dusky erythema, lymphedema, ulceration, vegetating plaques	Skin-colored, firm, mobile subcutaneous nodules
Size of lesions	0.2 to >5 cm	1–3 mm papules, annular plaques usually <6 cm	2 to >10 cm	3 to >10 cm	Variable	1–3 cm
No. of lesions	Variable	1–10	1–10	1–10	1–5	1–10
Associations	Systemic manifestations of sarcoidosis (see Table 93.3); can be drug-induced (e.g. IFN, TNF inhibitors) or reaction pattern to underlying lymphoma	Possible diabetes mellitus, thyroid disease, or hyperlipidemia; rare reports of HIV infection, malignancy	Actinic damage	Diabetes mellitus	Intestinal Crohn disease	Rheumatoid arthritis
Special clinical characteristics	Develop within sites of trauma including scars and tattoos	Central hyperpigmentation	Central atrophy and hypopigmentation	Yellow–brown atrophic centers, ulceration	Draining sinuses and fistulas	Occasional ulceration, especially at sites of trauma

Palisading granuloma

1. GA
2. NBL
3. AEGCG
4. RN
5. NECRBIOTIC XANTHOGRANULOMA

Palisading granuloma

	Granuloma annular	Necrobiosis libodica
epidemiolgy	More common in femal ,mainly <30yrs	Mainly in diabetic patient
pathogensis	unkown,trauma, insect bite reaction, sun exposur,borellia may be associated HLAB35	Immunologicaly mediated vascular disease. it has been postulated that the microangiopathic vessel changes seen in diabetic patients could contribute to the development of collagen degeneration and subsequent dermal inflammation.
c/p	Benign self limited granulomatous disease presented wih arciform or annular plaques at dorsum of hand , feet	yellow–brown, atrophic, telangiectatic plaques with an elevated violaceous rim, typically in the pretibial region
Clinical varient	macular ,papular ,perforating, sc Generalized mainly with hyperlipidemia ,localized	
Pathology	Focal degeneration of callagen ,elastic fibers , mucin deposition , perivascular and interstitial lymphocyte infiltrate in superficial mid dermis	Palisading granuloma “layered” tiers of granuloma surrounded by tiers of degenerated collagen with lipid deposition and sclerosis

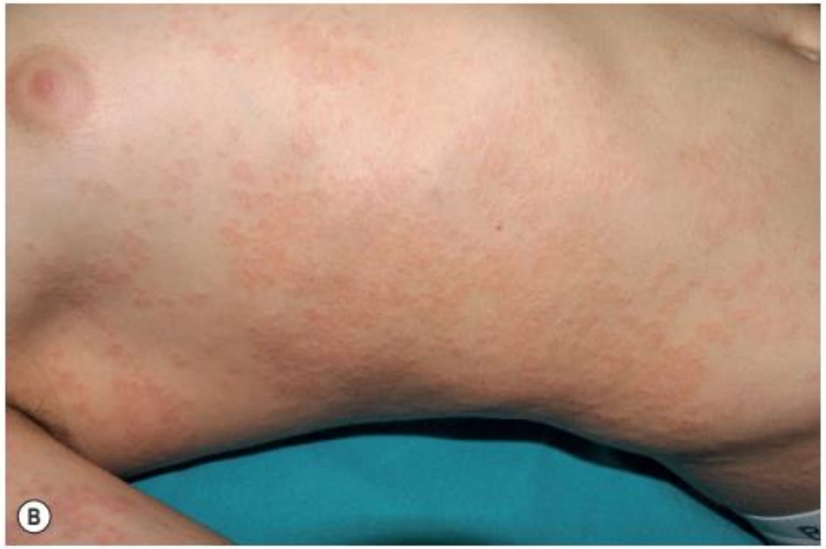
Palisading granuloma

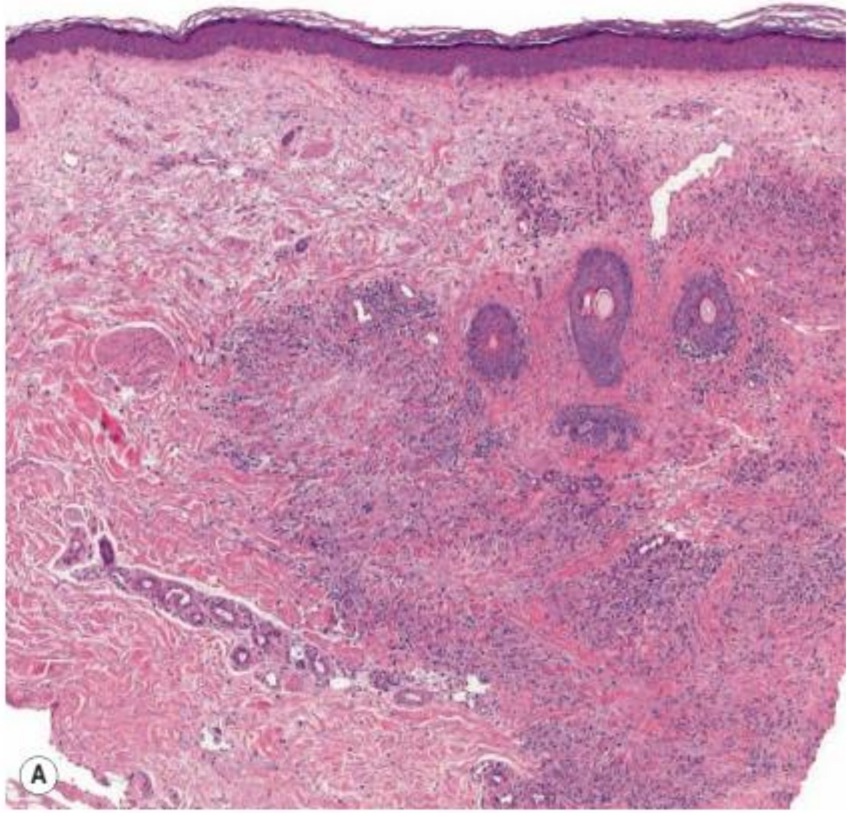
	Granuloma annular	Necrobiosis lipodica
comblication		Ulceration ,scc ,anesthia , alopecia ,hypohidrosis
DD	Annular lesions(annular sarcoidosis,AEGCG) , palisading granuloma	GA, NXG, sarcoidosis, diabetic dermopathy and lipodermatosclerosis. O
association	Autoimmune thyroiditis malignancy, HIV, hyperlibidemia	
treatment	Self limited ,reassurance , High potency topical steroid ,ILCS , phototherapy ,cryosurgery ,antimalarial.	No effective treatment even control diabetis. Potent topical steroid for early lesion. ILCS for active border .avoid trauma ,local care of ulcer.

Fig. 93.8 Granuloma annulare – clinical features. A,B Annular pink to pink–brown plaques; discrete papules may be seen. The dorsal aspect of the hand is a common location.

Fig. 93.9 Granuloma annulare – micropapular variant. Several of the lesions have a central dell. *Courtesy, Joyce Rico, MD.*

Fig. 93.10 Granuloma annulare – perforatin[g] and generalized/ disseminated variant[s] **A** Papules can have a central keratotic plug or umbilication. **B** Numerous small annular plaques on th[e] trunk of a child. *A, Court[esy,] Ronald P Rapini, MD; B, Court[esy,] Antonio Torrelo, MD.*





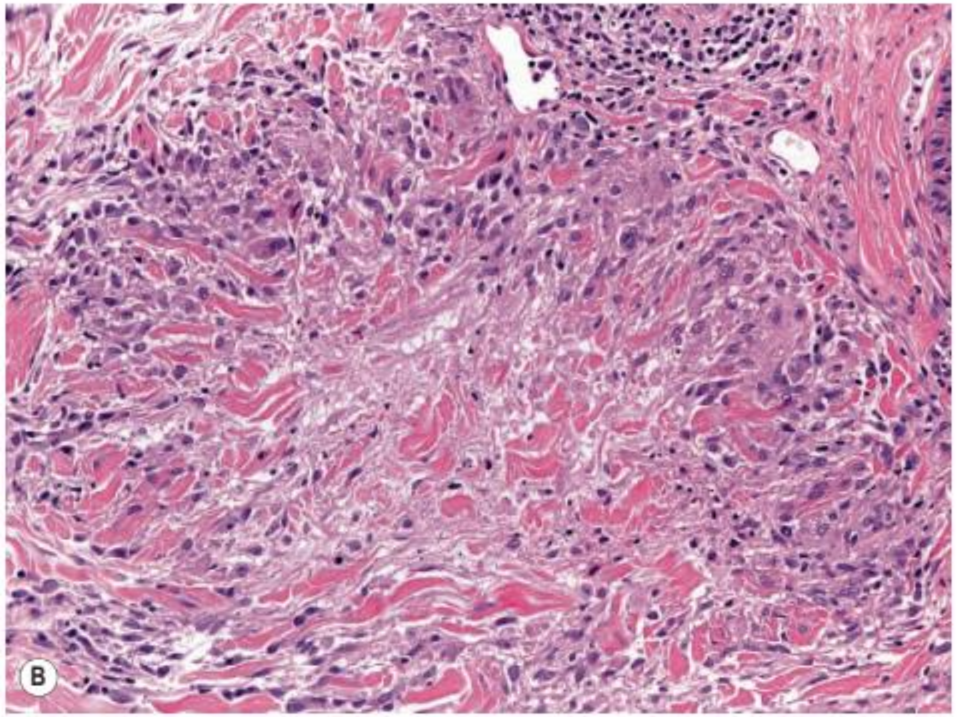


Fig. 93.12 Granuloma annulare – palisading granuloma pattern. A Multiple palisaded granulomas within the dermis. Degenerative collagen and mucin deposition are present in the central portion of the granulomas. **B** A palisaded granuloma with epithelioid histiocytes surrounding an anuclear dermis

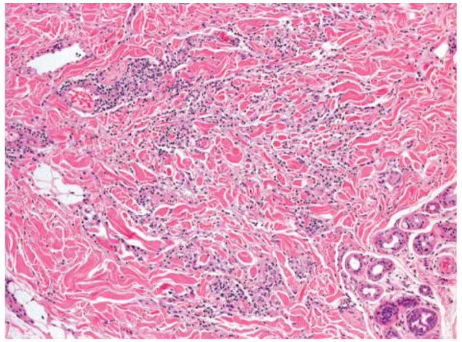


Fig. 93.11 Granuloma annulare – interstitial or infiltrative pattern. Infiltration of histiocytes between the dermal collagen fibers, with scant mucin. *Courtesy,*

TREATMENT OF GRANULOMA ANNULARE

Rx

Topical corticosteroids (3)
Intralesional corticosteroids (2)
Cryosurgery (2)
Topical calcineurin inhibitors (3)
Topical imiquimod (3)
Hydroxychloroquine (6 mg/kg/day) or chloroquine (3 mg/kg/day)^ (2)
Niacinamide (nicotinamide; 500 mg TID) (3)
PUVA (psoralen plus UVA) or UVA1 (2)
Pentoxifylline (3)
Intralesional interferon-γ (3)
5-lipoxygenase inhibitor (zileuton) plus vitamin E† (3)
Dapsone (100 mg/day) (3)
Isotretinoin (0.5–0.75 mg/kg/day), acetretin (3)
Minocycline + ofloxacin + rifampin* (3)
Methotrexate (3)
Prednisone (3)
Cyclosporine (3–4 mg/kg/day × 3 months) (3)
TNF-α inhibitors (adalimumab, infliximab) (3)
Fumaric acid esters (3)
Chlorambucil (3)
Photodynamic therapy with topical 5-aminolevulinic acid (3)
Lasers: CO_2, pulsed dye (585 nm), excimer (308 nm) (3)
Electrodesiccation (3)
Surgical excision (3)

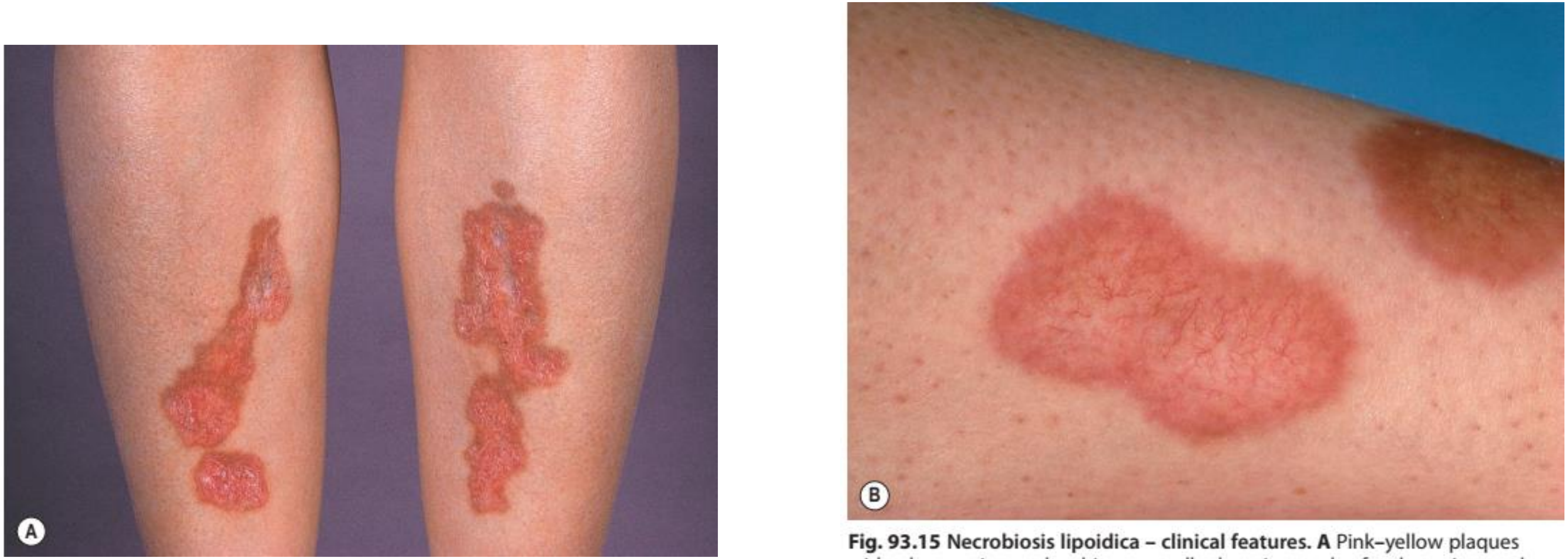


Fig. 93.15 Necrobiosis lipoidica – clinical features. A Pink–yellow plaques with a brown rim on the shins; centrally there is atrophy, focal scarring, and telangiectasias. **B** Annular pink–brown plaques with central telangiectasias and

eFig. 93.13 Necrobiosis lipoidica. Pink–brown plaque on the shin with central atrophy and subtle yellow color.

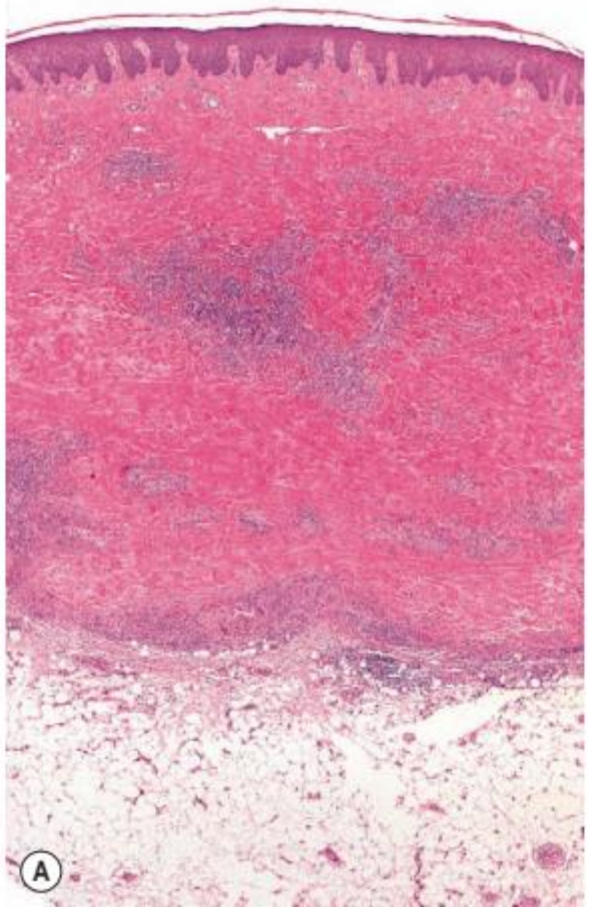
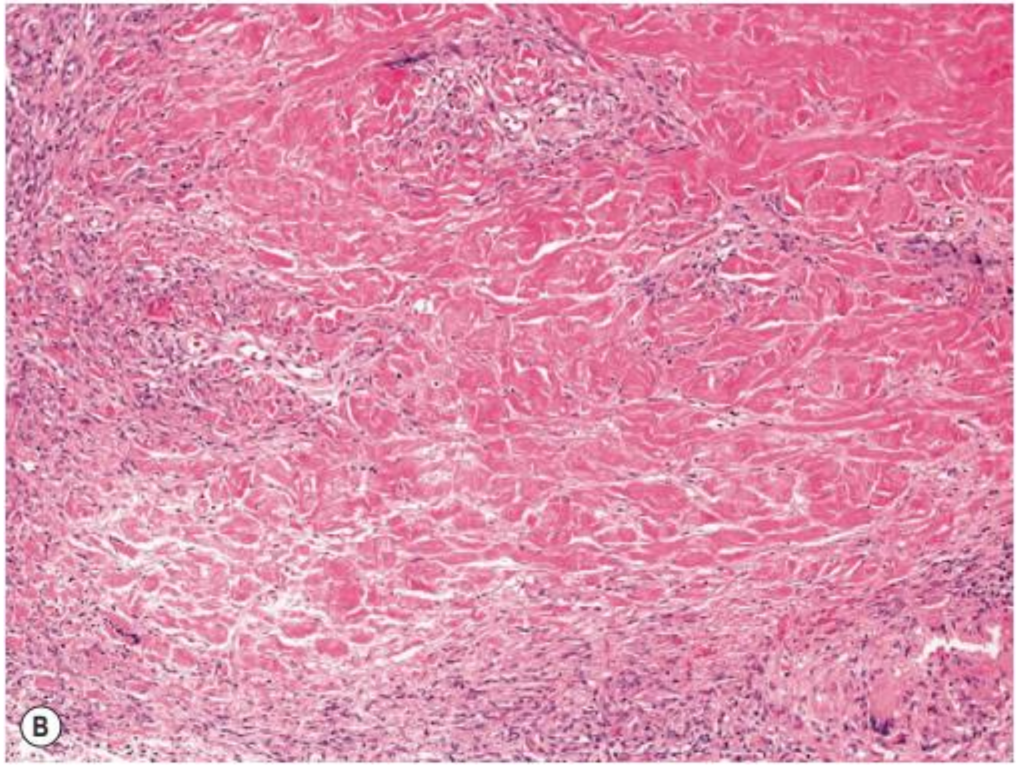
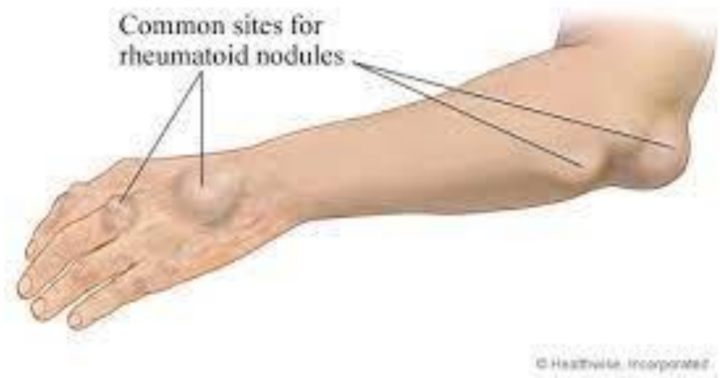


Fig. 93.16 Necrobiosis lipoidica – histopathologic features. A Multiple granulomas within the entire dermis extending into the subcutaneous fat. Note the layered tiers of granulomatous inflammation aligned parallel t the skin surface. **B** Palisaded granuloma composed of epithelioid histiocytes with altered collagen centrally.
Courtesy, Lorenzo Cerroni, MD.



	AEGCG "actinic granuloma"	REUMATOID NODULE
epidemiology	Uncommon disease mainly in middle aged women	
pathogensis	Inflammation precipitated by actinic damag	
C/P	medium-sized to large annular plaques with a raised erythematous border and a slightly atrophic, hypopigmented center, distributed mainly in chronically sunexposed areas "neck ,face"	Subcutaneus firm papulonodules periarticular over extensor surface
histopathology	in the raised border, there are -palisading granulomas composed of histiocyt foreign body type multinucleated giant cells that have engulfed elastic fibers (elastophagocytosis).	Palisadion granuloma in deep dermis ,sc

	AEGCG	RN
DD		
treatment	Topical intralesional corticosteroid , puva	



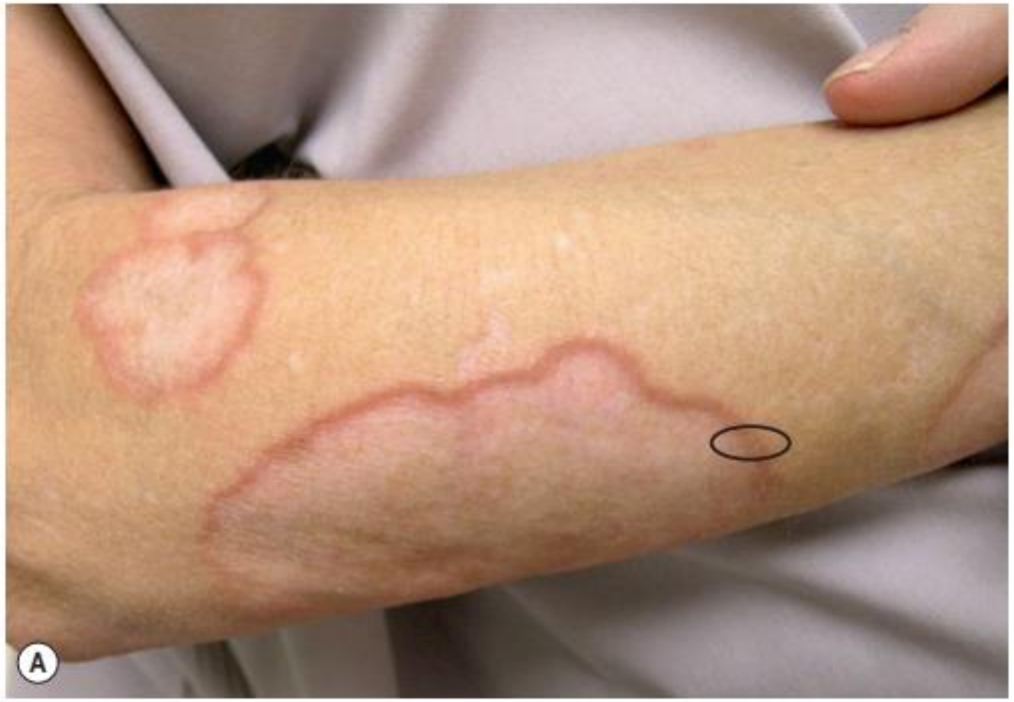


Fig. 93.13 Annular elastolytic giant cell granuloma – clinical features. A,B

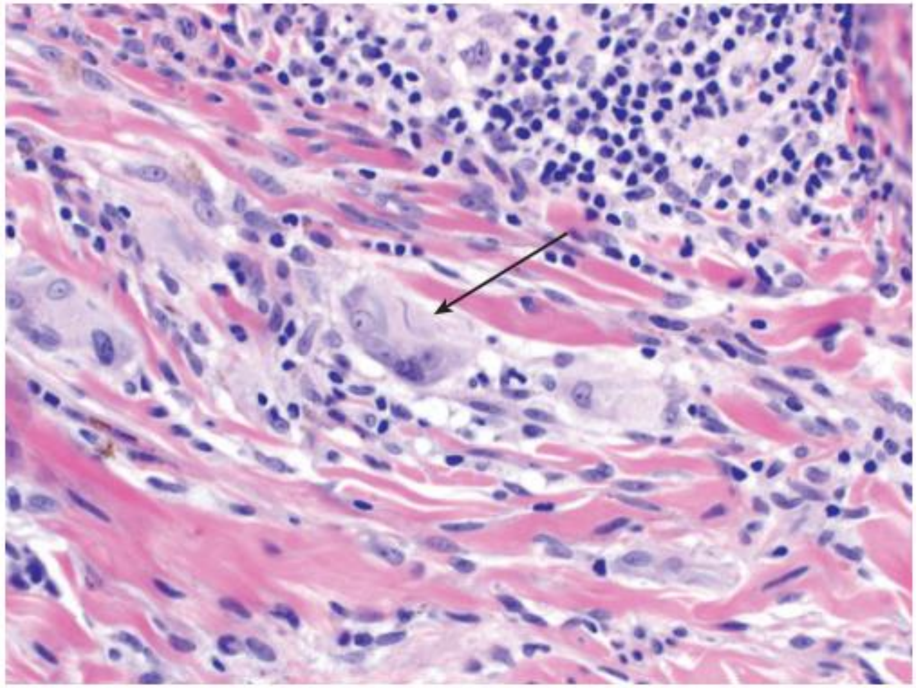
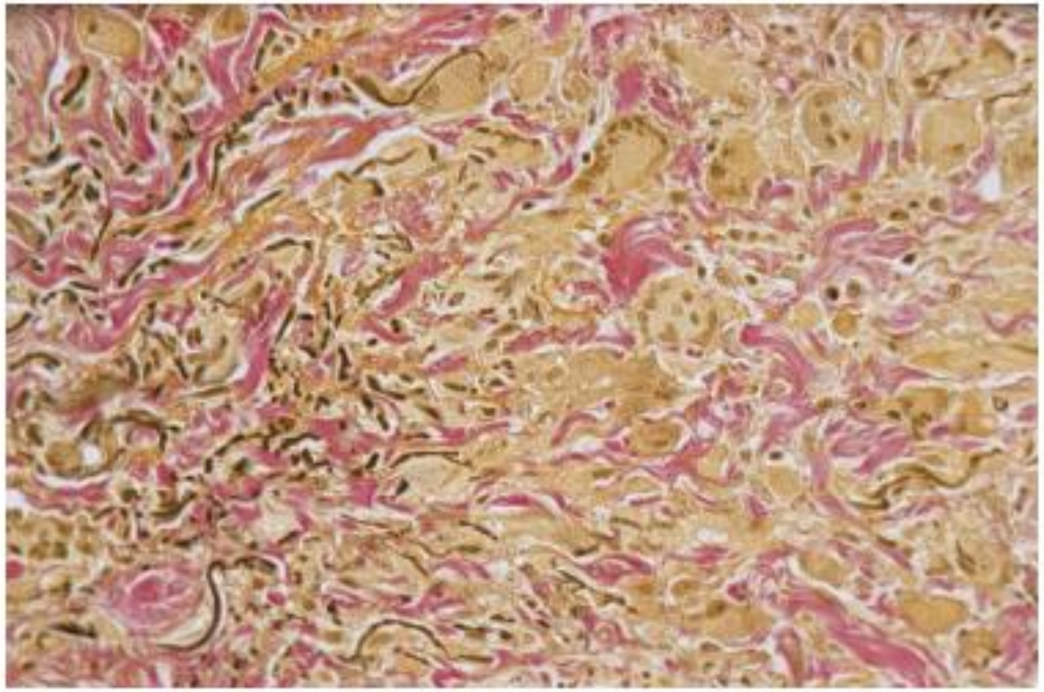
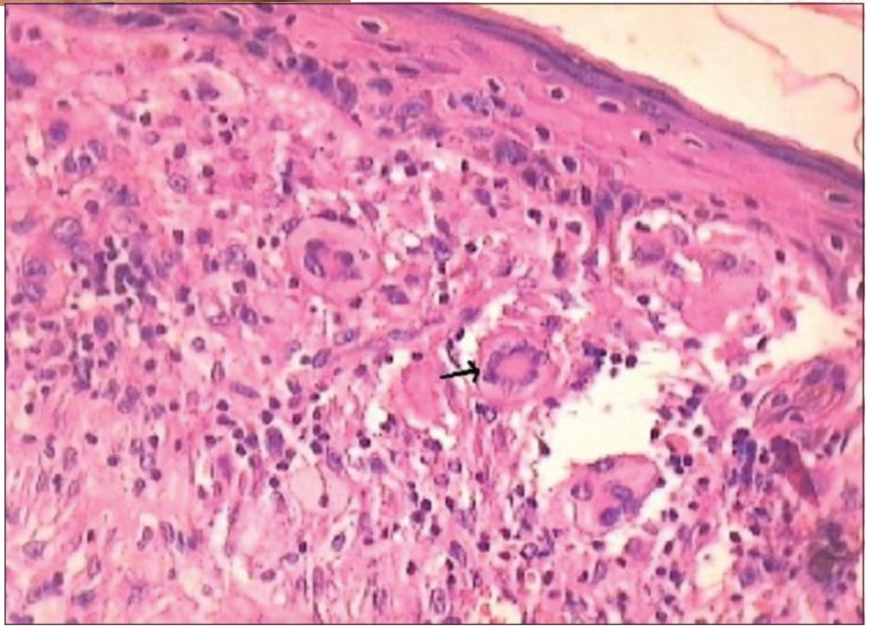


Fig. 93.14 Annular elastolytic giant cell granuloma – histopathological features. A dermal inflammatory infiltrate with several histiocytic giant cells. The presence of a fragmented elastic fiber within a giant cell



eFig. 93.12 Annular elastolytic giant cell granuloma – histologic features. Loss of elastic fibers (black) is seen amid the granulomatous infiltrate (yellow). The remaining collagen is stained red (Verhoeff–van Gieson stain). *Courtesy, Ronald P Rapini, MD.*

	Necrobiotic xanthogranuloma
	Progressive multisystemic non Langerhans cell histiocytosis →igG monoclonal gammopathy
c/p	• Destructive cutaneous , sc lesions • Cutaneous :multipleyellow papules , nodules →periorbital • Eye : 5%orbital masses →ectrobian , ant. Uveitis • Systemic : endocarditis , hepatosplenomegaly
h.P	• Palisading granuloma →touton bodies →foamy histeocyte • Necrobiosis, cholesterol cleft • Cd68 +ve
ttt	• Chlorambucil , melphalan , cyclophosphamide co2 laser • plasmapharesis



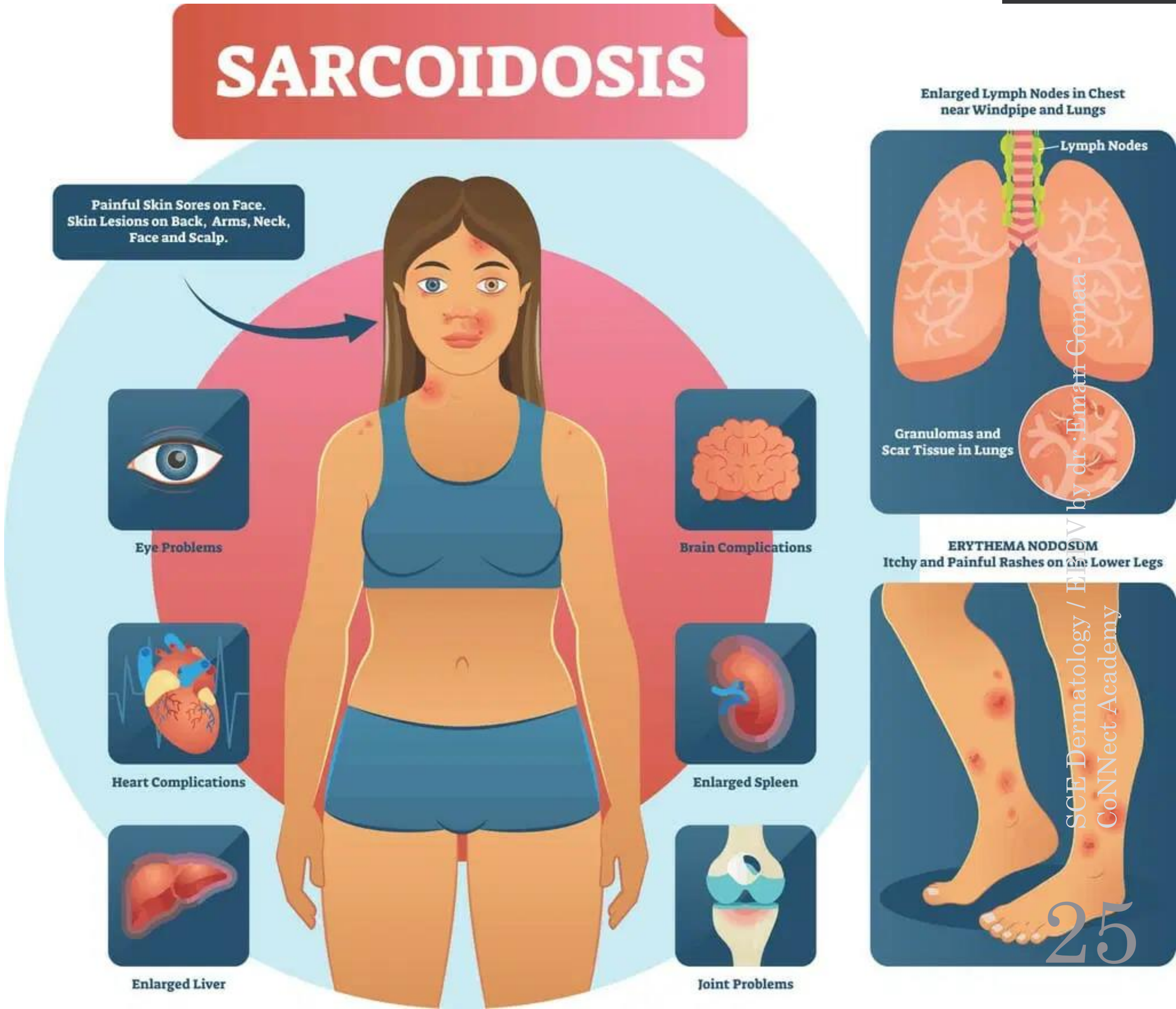
	Cutaneous CORHN disease
epidemiology	Crohn disease usually begins between the second and fourth decades of life. Mucocutaneous findings occur in 20–45% of patients.
pathogenesis	Cutaneous Crohn disease occurs in association with or precedes intestinal Crohn disease, reflecting a multisystem disease. - A number of genetic abnormalities which can lead to an exaggerated T-cell response to certain commensal enteric bacteria and defective microbial clearance have been linked to Crohn disease.
C/P	Specific lesions 1-GENITAL LESION: Labial, penile or scrotal erythema and/or lymphedema are common presenting signs and Linear or “knife-like” ulcerations that are within body folds , vegetative plaques and skin tags. 2- ORAL LESION: occur in 5–20% of patients with Crohn disease, and these may consist of cobblestoning of the buccal mucosa, tiny gingival nodules, gingival hyperplasia, aphthae-like ulcers , linear, knife-like ulcerations, pyostomatitis vegetans. 3_SKIN LESION: nodules, gingival hyperplasia, aphthae-like ulcers (Fig. 93.17F), linear, knife-like ulcerations, pyostomatitis vegetans

	Cutaneous corhn disease
C/P	NON SPECIFIC LESIONS: reactive cutaneous manifestations: polyarteritis nodosa, erythema nodosum, erythema multiforme, finger clubbing, cutaneous small vessel vasculitis, epidermolysis bullosa acquisita, vitiligo, palmar erythema.
pathology	In both cutaneous and oral lesions of Crohn disease, nodular, noncaseating, epithelioid tubercles with surrounding lymphocytes are found in the superficial and deep dermis, sometimes extending into the subcutaneous fat.
DD	other granulomatous disorders such as cutaneous sarcoidosis, mycobacterial infections, deep fungal infections, and foreign body reactions. Other infections, such as actinomycosis or cellulitis.
TREATMENT	• topical or intralesional corticosteroids and/or topical calcineurin inhibitors . Oral metronidazole (250 mg TID for at least 4 months) can be an effective treatment. • systemic medications including corticosteroids, sulfasalazine, azathioprine, 6-mercaptopurine, TNF-α inhibitors (e.g. infliximab, adalimumab), and thalidomide may lead to improvment

Fig. 93.17 Cutaneous Crohn disease. A Vulvar and perianal erythema and induration. Note the marked asymmetric vulvar swelling and perianal erosions. **B** Marked lymphedema of the prepuce with lymphedema and induration of the scrotum. **C** Inflammation and lymphedema leading to twisting of the penis along its long axis, referred to as a "saxophone penis". **D** Perianal tags and indurated plaques. **E** Firm erythematous plaques of the mons pubis and labia majora and minora. There are also draining sinuses. **F** Intraoral ulcer that resembles a major aphthous ulcer. **G** Ulcerated erythematous plaque of the mandible with draining sinuses. *A, Courtesy, Julie V Schaffer, MD; C,E,G, Courtesy, Luis Requena, MD; D, Courtesy, Mary Stone, MD.*

sarcoidosis

- Definition:
- systemicgranulomatous disorder of unknown origin that most commonly involves the lungs , skin ,kidney,eye other organs



Enviromental

Genatic

immunological

Affect organ exposed to environment

- Pollen
- Insectside
- Moulds
- tb
- Borellia
- Hhv 8
- ebv

HLA B1
HLA DQ B1
HLADRB1

CMI →Th1

Fig. 93.2 Cutaneous sarcoidosis – papules and plaques. **A** Cutaneous sarcoidosis usually consists of papules and plaques with a typical reddish-brown to violet-brown color. **B, C** Lesions often favor the nose, lips and perioral region. **D** Hyperpigmented plaques, some of which have scale. **E** Papules of cutaneous sarcoidosis arising within a tattoo; the differential diagnosis includes foreign body

- sarcoidosis
  - cutaneouse
  - systemic
  - syndrom
  - associations

ctaneouseu

- Specific
  Sarcoidal granuloma
  - typical
  - atypical
- Non specific
  Reactive process
  - atypical

- Up to a third of patients with systemic sarcoidosis develop skin lesions, which may be the first or only clinical manifestation of the disease.

Typical lesion of sarcoidosis

- Lesions: asymptomatic multiple red-brown or red-violaceous papules and plaques.
- The overlying epidermis is usually normal (but occasionally scale or ulcerations can develop).
- Common sites: nose , face, lips, neck, upper trunk and extremities .
- Distribution: symmetrical.
- Diascopy (in which pressure induces blanching): the lesions show a subtle brown-yellow or "apple jelly" color characteristic of granulomatous .

- Variants: Maculo-papular sarcoidosis , Nodular and plaque sarcoidosis , Lupus pernio , Angiolupoid sarcoidosis , Scar sarcoidosis , Subcutaneous sarcoidosis(darier roussy) .

Fig. 93.2 Cutaneous sarcoidosis – papules and plaques. A Cutaneous sarcoidosis usually consists of papules and plaques with a typical reddish-brown to violet–brown color. **B, C** Lesions often favor the nose, lips and perioral region. **D** Hyperpigmented plaques, some of which have scale. **E** Papules of cutaneous sarcoidosis arising within a tattoo; the differential diagnosis includes foreign body

ATYPICAL SAROCOIDAL LESION

- Hypopigmented , Lichenoid , Ulcerative
- Psoriasiform , Verrucous , Ichthyosiform
- Erythrodermic , Morphoea-Iike sarcoidosis
- Livedo sarcoidosis

Non-specific lesions

- Erythema nodosum (EN): the most common (good prognosis).
- Others less common non-specific lesions: erythema multiforme, prurigo, cutaneous calcinosis and digital clubbing (a poor prognostic sign).

Fig. 93.3 Cutaneous sarcoidosis – clinical variants. A The hypopigmented variant is more noticeable in individuals with darkly pigmented skin. **B** Ichthyosiform presentation with obvious scale. **C** Coalescing violaceous papules on the nose in lupus pernio; note the notching of the nasal rim. *A, Courtesy, Louis A Fragola, Jr, MD; B, Courtesy, Jean L Bolognia, MD.*

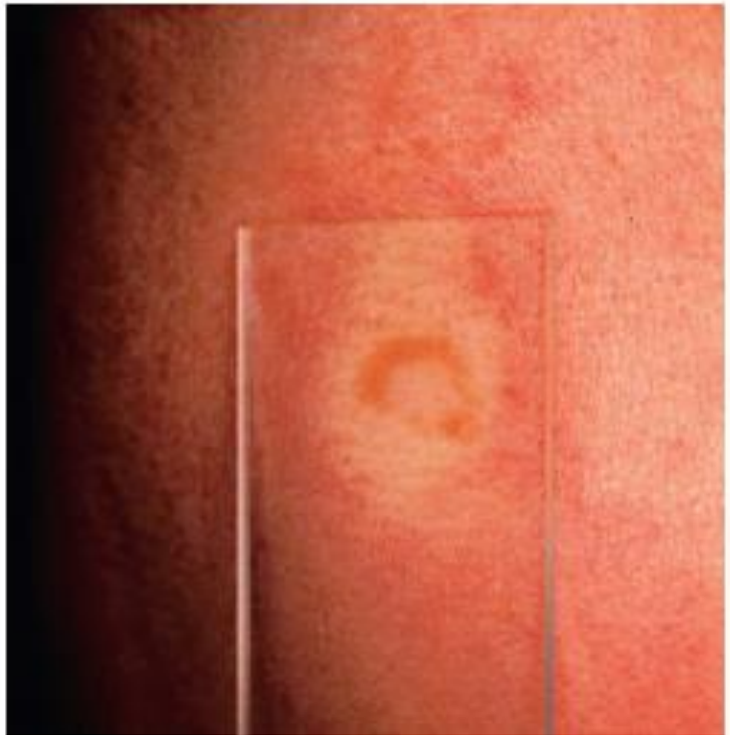


Fig. 93.4 Cutaneous sarcoidosis – diascopy. With diascopy, a yellow–brown, "apple jelly" color is seen.

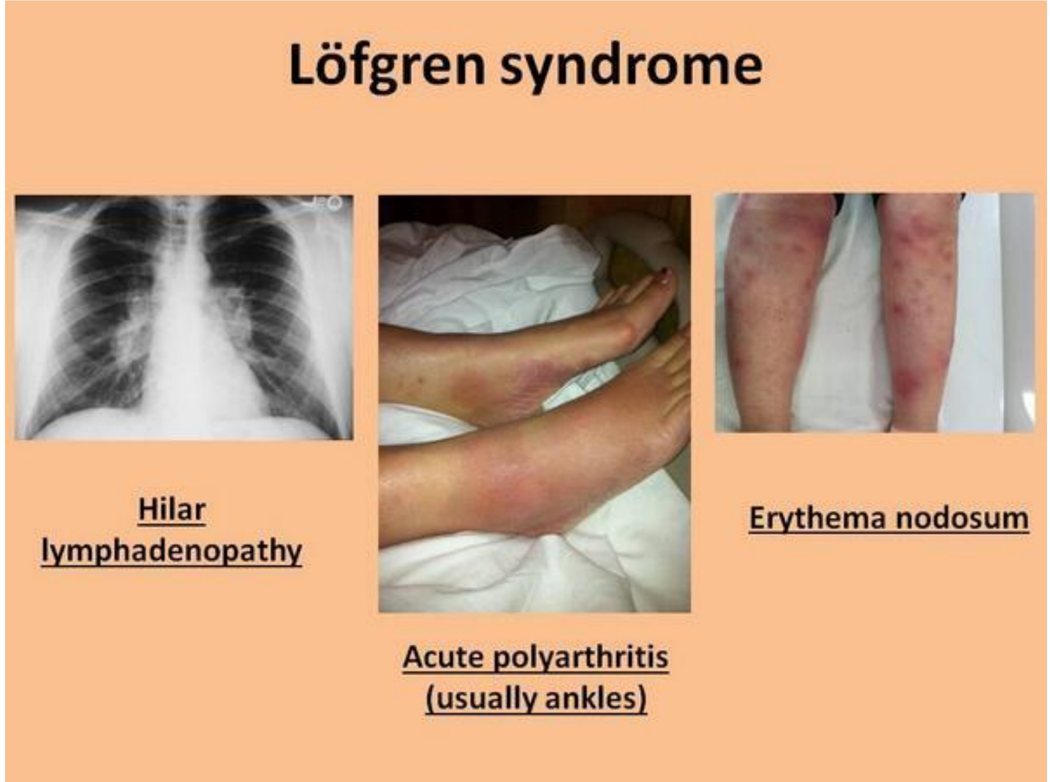
Fig. 93.5 Erythema nodosum in a patient with sarcoidosis. These patients often have hilar lymphadenopathy. Löfgren syndrome consists of erythema

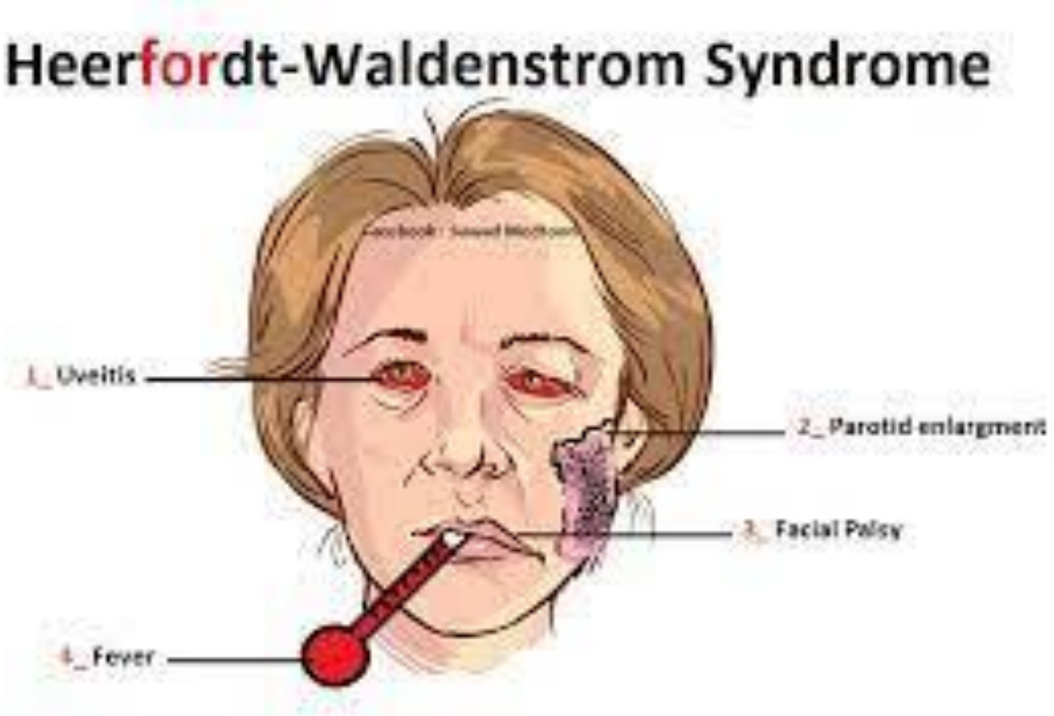
SYSTEMIC MANIFESTATION

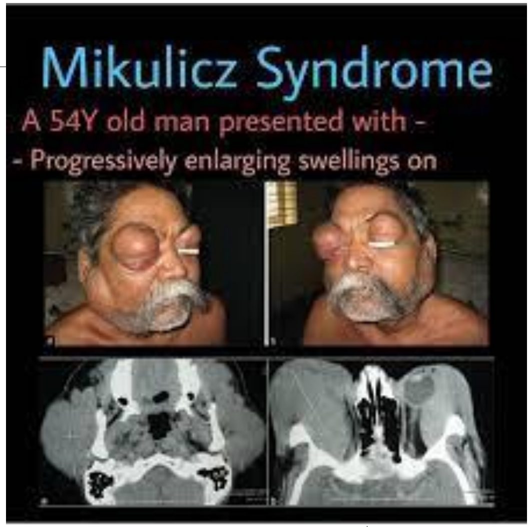
organ	%of patient	C/P	evaluation
lung	90–95%	Dyspnea, non-productive cough/pulmonary infiltrates, fibrosis, restrictive lung disease	CXR , chest ct scan
Lymph nodes (LN)	30–40*	Lymphadenopathy/enlarged hilar and/or paratracheal LN	CXR, high resolution chest CT scan.
Eyes	25%	Uveitis (anterior) (can be asymptomatic despite being severe), conjunctivitis, sicca symptoms	Yearly ophthalmologic examination
Kidneys	10–40	Nephrolithiasis/hypercalciuria, ↓renal function	BUN, crt, serum calcium, spot urine calcium : crt ratio,
Liver/spleen	10–20%	Hepatomegaly and/or splenomegaly (rarely clinically relevant), cirrhosis, consequences of splenic enlargement (e.g. thrombocytopenia)/↑LFTs, ↓plts	• LFTs, physical examination • If clinically relevant hypersplenism suspected, abdominal/pelvic CT scan

Endocrine	5–10 %	• Pituitary or thyroid dysfunction • Hypercalcemia (increased calcitriol synthesis by sarcoidal histiocytes)	• Thyroid function test • Expanded hormonal testing when clinically indicate
Heart	25 (5% clinically relevant)	Palpitations, sudden death, CHF/arrhythmias, cardiomegaly	echocardiogram, (PET scan, cardiac MRI)
Upper respiratory tract	5–10 %	Sinusitus, nasal congestion, stridor, parotiditis	Referral to otolaryngologist
Bone marrow	50%	Lymphopenia , leukopenia, eosinophilia, hypergammaglobulinemia	cbc
CNS and peripheral nervous system	10–20%	Neuropathies – cranial, spinal cord, peripheral, small fiber	MRI, nerve conduction studies

Syndroms	associations
(1) Lofgren syndrome: EN, bilateral hilar lymphadenopathy (with or without pulmonary infiltrates), fever, migrating polyarthritis, and acute iritis (anterior uveitis). Prognosis is very good and it usually resolves spontaneously within 1 year.	• Autoimmune diseases: Sjogren syndrome, systemic sclerosis, rheumatoid arthritis, vasculitis, psoriasis, autoimmune chronic hepatiti
(2) **Heerfordt syndrome** (uveo-parotid fever): uveitis, parotid enlargement, facial nerve palsy, and fever	• Malignancies: lymphoproliferative (mainly Hodgkin lymphoma) and solid tumors
(3**) Mikulicz syndrome:** bilateral enlargement of the parotid, sublingual, lacrimal, or submandibular glands	


Löfgren syndrome
Hilar lymphadenopathy
Acute polyarthritis (usually ankles)
Erythema nodosum


Heerfordt-Waldenstrom Syndrome
1_ Uveitis
2_ Parotid enlargment
3_ Facial Palsy
4_ Fever


Mikulicz Syndrome
A 54Y old man presented with -
- Progressively enlarging swellings on

- Pathology:

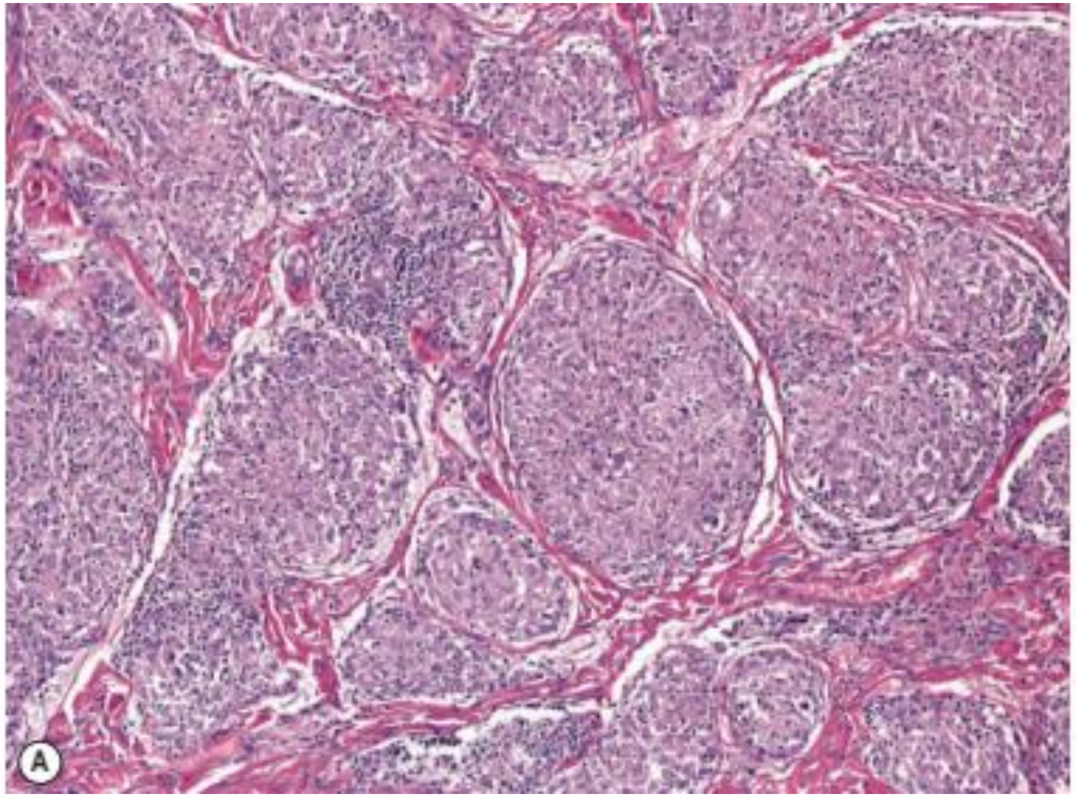
The histopathologic hallmark of sarcoidosis is the presence of superficial and deep dermal epithelioid cell granulomas devoid of prominent infiltrates of lymphocytes or plasma cells (“naked tubercles”) Central caseation is usually absent.

(3) Investigations to assess disease activity:

- Angiotensin converting enzyme (ACE)

The serum ACE level is elevated in -60% of patients (produced by sarcoidal granuloma).

- Other markers of disease activity:
- ・INF-a, IL-2 —» markers for disease activity
- ・IL-8, IL-6 —► with granulomatous lung disease



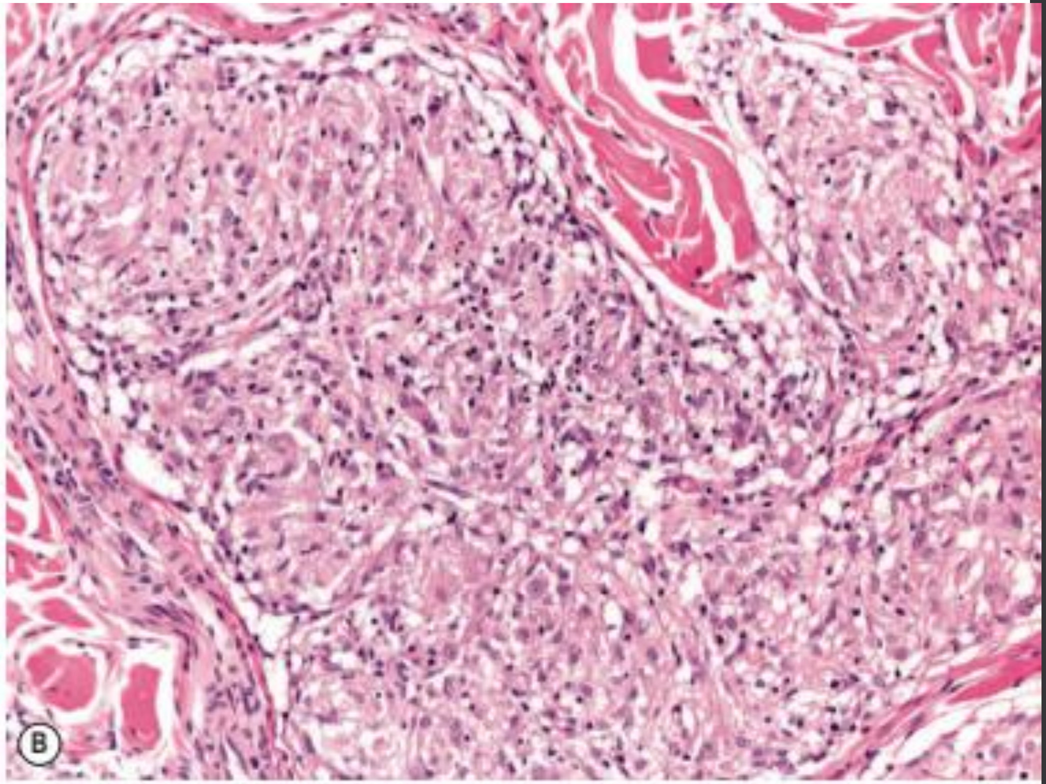


Fig. 93.6 Cutaneous sarcoidosis – histopathologic features. A Nodular aggregates of epithelioid histiocytes forming tubercles filling the dermis. **B** A sarcoidal tubercle with a sparse admixture of lymphocytes ("naked tubercle").

TREATMENT OF CUTANEOUS SARCOIDOSIS

- Topical or intralesional corticosteroids
- Topical calcineurin inhibitors
- Oral tetracyclines (e.g. minocycline, doxycycline)
- Oral hydroxychloroquine or chloroquine
- Methotrexate
- Systemic corticosteroids
- Thalidomide
- TNF-α inhibitors (adalimumab, infliximab)
- Intralesional chloroquine
- Allopurinol
- Isotretinoin
- PUVA (psoralen plus UVA)
- Mycophenolate mofetil
- Pulsed dye or CO_2 laser
- Photodynamic therapy
- Surgical excision

HISTOLOGIC FEATURES OF THE MAJOR NON-INFECTIOUS GRANULOMATOUS DERMATITIDES

	Sarcoidosis	Granuloma annulare	Necrobiosis lipoidica	AEGCG	Cutaneous Crohn disease	Interstitial granulomatous dermatitis	Palisading neutrophilic and granulomatous dermatitis	Rheumatoid nodule*
Typical location	Superficial and deep dermis†	Superficial and mid dermis†	Entire dermis, subcutis	Superficial and mid dermis	Superficial and deep dermis	Mid and deep dermis	Entire dermis	Deep dermis, subcutis
Granuloma pattern	Tubercle with few peripheral lymphocytes ("naked")	Palisading or interstitial; patchy with discrete foci	Diffuse palisading and interstitial**; horizontal "tiers"	Palisading, irregular	Tubercle with surrounding lymphocytes	Interstitial and small "rosettes" around single collagen fibers	Palisading and interstitial; prominent neutrophils and leukocytoclasia	Palisading
Necrobiosis (altered collagen)	No	Yes ("blue")	Yes ("red")	No	No	Yes ("blue")	Yes ("blue")	Yes ("red")
Giant cells	Yes	Variable	Yes	Yes	Yes	Variable	Variable	Yes
Elastolysis	No	Variable	Variable	Yes	No	Variable	Variable	No
Elastophagocytosis	No	Variable	No	Yes	No	No	No	No
Asteroid bodies	Yes	Variable	Variable	Yes	No	Variable	Variable	No
Mucin	No	Yes#	Minimal	No	No	Minimal	Variable	Variable
Extracellular lipid	No	Variable	Yes	No	No	No	No	Variable
Vascular changes	No	Variable	Yes	No	No	No	Yes	Yes


Thank you